POWER-TORSION EXT GROUPS ARE ESSENTIALLY CONSTANT ALONG THE QUOTIENTS BY POWERS OF THE IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes 'is essentially constant' in the statement display and 'A diagram chase shows' in the proof; both are reproduced unchanged. The single reference inside the counted proof span is recorded as a non-load-bearing navigation pointer.

Allowed standard prerequisites (editorial).

- Finite generation of Ext^i_A(K, M) for pseudo-coherent K and finite M. used at proof lines 29796-29798. Makes the power-torsion hypothesis effective, giving the annihilator I^t.
- Vanishing of the image of Ext^i_A(K, I^nM) -> Ext^i_A(K, M) for large n. invoked at proof lines 29799-29801. Is the annihilation input that the long exact sequence consumes on both sides.
- The long exact sequence of Ext attached to a short exact sequence of modules. used at proof lines 29802-29807. Produces the four-term sequence from which essential constancy is deduced.

Source reference. Original lemma 29584: sources/stacks-more-algebra/pinned-original-source.tex, source lines 29584–29584. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let A be a Noetherian ring. Let $I \subset A$ be an ideal. Let $K \in D(A)$ be pseudo-coherent. Let $a \in \mathbf{Z}$. Assume that for every finite A-module M the modules $\mathrm{Ext}^i_A(K, M)$ are I-power torsion for $i \geq a$. Then for $i \geq a$ and M finite the system $\mathrm{Ext}^i_A(K, M/I^nM)$ is essentially constant with value*

$$\mathrm{Ext}^i_A(K, M) = \lim \mathrm{Ext}^i_A(K, M/I^nM)$$

Proof. Let M be a finite A-module. Since K is pseudo-coherent we see that $\mathrm{Ext}^i_A(K, M)$ is a finite A-module. Thus for $i \geq a$ it is annihilated by I^t for some $t \geq 0$. By Lemma 29584 we see that the image of $\mathrm{Ext}^i_A(K, I^nM) \to \mathrm{Ext}^i_A(K, M)$

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is zero for some $n > 0$. The short exact sequence $0 \to I^n M \to M \to M/I^n M \to 0$ gives a long exact sequence

$$\mathrm{Ext}^i_A(K, I^n M) \to \mathrm{Ext}^i_A(K, M) \to \mathrm{Ext}^i_A(K, M/I^n M) \to \mathrm{Ext}^{i+1}_A(K, I^n M)$$

The systems $\mathrm{Ext}^i_A(K, I^n M)$ and $\mathrm{Ext}^{i+1}_A(K, I^n M)$ are essentially constant with value 0 by what we just said (applied to the finite A-modules $I^m M$). A diagram chase shows $\mathrm{Ext}^i_A(K, M/I^n M)$ is essentially constant with value $\mathrm{Ext}^i_A(K, M)$. $\square$